JUHO JEON

Integrated Circuit Design Engineer

(+852) 6893-2994 | eejjkr@gmail.com | linkedin.com/in/juho-jeon | github.com/eejuhojeon

EDUCATION

Bachelor of Engineering in Electronic Engineering Expected Jun 2024
Minor in Information Technology
The Hong Kong University of Science and Technology (HKUST)

- Relevant courses: ELEC5280 (High Frequency Circuit Design) ELEC 4430 (Integrated Power Electronics) ELEC4320 (FPGA-based Design) ELEC4420 (Analog Integrated Circuits Design and Analysis) ELEC4410 (CMOS VLSI Design)
- Received a Dean's List in spring 2023 for academic excellence.
- Received an offer for a Master of Philosophy program in Electronic Engineering 2024 Fall intake from Professor Patrick Yue

WORK EXPERIENCES

Cybersecurity Analyst Intern Feb 2022 – Dec 2022
Deloitte Advisory Ltd.

- Contribute to Identity and Access Management (IAM) engagements and bidding proposals to design and implement target processes of user management.

Electronic Engineering Intern Dec 2021 – Jan 2022
Gense Technologies Ltd.

- Support various projects in hardware development for an innovative portable medical device.

PROJECTS

Low Noise Amplifier Spring 2024 – Present
ELEC5280 (High Frequency Circuit Design) Project

- Built a low noise amplifier for a radio communication purpose using 22nm technology that performs at 28GHz center frequency, 16dB gain, 2.25dB NF, and 20mW power consumption.

Successive Approximation Register Analog-to-Digital Converter (SAR ADC) Jun 2023 – May 2024
Final Year Thesis Supervised by Professor Yihan Zhang

- Built a SAR ADC using 180nm technology that performs at a 2MS/s sampling rate, 10-bit resolution, and 9.09 ENOB, 1.92mW power consumption using a monotonic switching procedure.

Two Stage Differential Input Single-Ended Output Operational Amplifier Fall 2023
ELEC4420 (Analog Integrated Circuits Design and Analysis) Project

- Built a two-stage operational amplifier using 180nm technology that performs at 1.8V supply voltage, 55.7dB gain, 348MHz unity-gain frequency, 1.58 V_{pp} output voltage swing, and 1.79mW power consumption.

Full Adder Fall 2023
ELEC4410 (CMOS VLSI Design) Project

- Built a 4-bit Full Adder using 180nm technology and delivered DRC and LVS clean layout. The design performs at 1.8V supply voltage, 1.37ns propagation delay, 531um^2 area, and 2.87pW power consumption.

Electronic Music Box Spring 2023
STM32 based ELEC3400 (Embedded System) Project

- Created a music box that can play music with various sound effects such as low pass filter, high pass filter, volume control, pitch control, tempo control, and timelapse with an interactive display.

EXTRA-CURRICULAR ACTIVITIES

Electronic and Computer Engineering (ECE) Ambassador Sep 2023 – Present
Undergraduate Staff-Student Liaison Committee

Hardware Engineer Sep 2021 – Dec 2021
HKUST Robotics Team

Guitarist Sep 2017 – Jun 2019
Band Society at the Hong Kong University of Science and Technology

SKILLS & CAPABILITIES

Languages: Korean (Native), English (Proficient), Mandarin (Intermediate)
Engineering: Cadence Virtuoso, Verilog, VHDL, Python, C++, C, MATLAB, PCB Design